

Task 1: Create a Lambda function

4. In the search box to the right of **Services**, search for and choose **Lambda** to open the AWS Lambda console.
5. Choose **Create a function**.
6. In the **Create function** screen, configure these settings:
 - Choose **Author from scratch**
 - Function name: `myStopinator`
 - Runtime: **Python 3.11**
 - Choose **Change default execution role**
 - Execution role: **Use an existing role**
 - Existing role: From the dropdown list, choose **myStopinatorRole**
7. Choose **Create function**.

Task 2: Configure the trigger

In this task, you will configure a scheduled event to trigger the Lambda function by setting an Amazon EventBridge event as the event source (or trigger). The Lambda function can be configured to operate much like a cron job on a Linux server, or a scheduled task on a Microsoft Windows server. However, you do not need to have a server running to host it.

8. Choose **Add trigger**.
9. Choose the **Select a trigger** dropdown menu, and select **EventBridge (CloudWatch Events)**.
10. For the rule, choose **Create a new rule** and configure these settings:
 - Rule name: `everyMinute`
 - Rule type: **Schedule expression**
 - Schedule expression: `rate(1 minute)`

Note: A more realistic, schedule-based stopinator Lambda function would probably be triggered by using a cron expression instead of a rate expression. However, for the purposes of this activity, using a rate expression ensures that the Lambda function will be triggered soon enough that you can see the results.

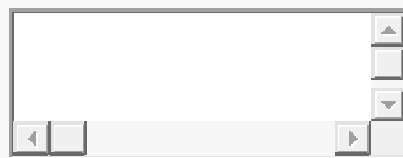
11. Choose **Add**.

Task 3: Configure the Lambda function

In this task, you will paste a few lines of code to update two values in the function code. You do not need to write code to complete this task.

12. Below the **Function overview** pane, choose **Code**, and then choose *lambda_function.py* to display and edit the Lambda function code.

13. In the **Code source** pane, delete the existing code. Copy the following code, and paste it in the box:



```
import boto3
region = '<REPLACE_WITH_REGION>'
instances = ['<REPLACE_WITH_INSTANCE_ID>']
ec2 = boto3.client('ec2', region_name=region)

def lambda_handler(event, context):
    ec2.stop_instances(InstanceIds=instances)
    print('stopped your instances: ' + str(instances))
```

Note: After pasting the code into the **Code source** box, review line 5. If a period (.) was added, delete it.

14. Replace the `<REPLACE_WITH_REGION>` placeholder with the actual Region that you are using. To do this:

Choose on the region on the top right corner and use the region code. For example, the region code for US East (N. Virginia) is *us-east-1*.

Important: Keep the single quotation marks (' ') around the Region in your code. For example, for the N. Virginia, it would be `'us-east-1'`

15. **Challenge section:** Verify that an EC2 instance named *instance1* is running in your account, and copy the *instance1* **instance ID**.

You are encouraged to figure out how to do this task without specific step-by-step guidance. However, if you need detailed guidance, select this text to reveal detailed steps:

16. Return to the **AWS Lambda console** browser tab, and replace `<REPLACE_WITH_INSTANCE_ID>` with the actual instance ID that you just copied.

Important: Keep the single quotation marks (' ') around the instance ID in your code.

Your code should now look similar to the following example. However, you might have a different value for the Region, and you will have a different value for the instance ID:

17. Choose the **File** menu and **Save** the changes. Then, in the **Code source** box, choose **Deploy**.

Your Lambda function is now fully configured. It should attempt to stop your instance every minute.

18. Choose **Monitor** (the tab near the top of the page).

Note that one of the charts shows you how many times your function has been invoked. There is also a chart that shows the error count and the success rate as a percentage.

Task 4: Verify that the Lambda function worked

19. Return to the **Amazon EC2 console** browser tab and see if your instance was stopped.

Tip: You can choose the refresh icon or refresh the browser page to see the change in state more quickly.

20. Try starting the instance again. What do you think will happen?

Choose **here** to reveal the answer.